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Simultaneous Equations

Two equations, two unknowns. Both linear? Eliminate. One has a square? Substitute. Calculator first, always.

THE TOPIC ON ONE MAP

Simultaneous Equations

01 Linear: elimination

$$3x + 5y = 6, \quad 7x - 5y = -11$$

$$\text{decimals: } 3x + 5y = 3.1$$

letters a, c

02 Forming them from words

apples and pears

tickets

ratio, sequence

03 Graphs: the crossing point

"by drawing another line"

estimate from a grid

04 Linear and quadratic: substitution

$$x^2 + y^2 = 41, \quad 2x + y = 3$$

"intersect at A and B"

05 Show that: tangent, one point, midpoint

"is a tangent"

"exactly one point"

midpoint of PQ

ZONE A

Both linear

Eliminate (Idea 01)

ZONE B

One has x^2 or xy

Substitute (Idea 04)

ZONE C

Words

Name the letters (Idea 02)

ZONE D

A graph

Read the crossing (Idea 03)

01 Linear Pairs: Elimination

Get the answer on the calculator first. Then make one letter match, and remove it: **Same Sign → Subtract, Different Sign → Add.**

WHEN YOU SEE → YOU DO

two linear equations

→ calculator first (MODE 5 1), then eliminate

+5y and -5y (different signs)

→ ADD

6y and 6y (same signs)

→ SUBTRACT

nothing matches

→ multiply one or both equations first

Calculator

Match

Add or Subtract

Solve

Substitute back

WORKED MODEL • ELIMINATION

Solve $5x + 2y = 11$ and $4x - 3y = 18$.

1 calculator: $x = 3, y = -2$

MODE 5 1

2 $15x + 6y = 33$ and $8x - 6y = 36$

match: $\times 3, \times 2$

3 $23x = 69 \Rightarrow x = 3$

different signs: add

4 $5(3) + 2y = 11 \Rightarrow 2y = -4$

substitute back

ANSWER $x = 3, y = -2$

× **TRAP** Subtracting a negative: $6y - (-6y) = 12y$. Bracket the second equation when you subtract.

IDEA 01

Practise: Linear Pairs: Elimination

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Different signs: add

CH 2 · TOPIC 12 3 marks

Solve $3x + 2y = 16$ and $5x - 2y = 8$

Q2 **DR ESLAM** Match first

CH 2 · TOPIC 12 3 marks

Solve $2x + 3y = 12$ and $5x + 2y = 19$

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 12

3 marks

1 Solve the simultaneous equations

$$4x + y = 25$$

$$x - 3y = 16$$

(3 marks)

Q4**PRACTICE**

Practice · Easy

CH 2 · TOPIC 12

3 marks

7 Solve the simultaneous equations

$$x + 2y = -0.5$$

$$3x - y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(3 marks)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 12

4 marks**4** Solve the simultaneous equations

$$4x + 7y = 1$$

$$3x + 10y = 15$$

(4 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12 5 marks

15 Given that

$$m \begin{pmatrix} 4 \\ 1 \end{pmatrix} + n \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

find the value of m and the value of n . $m = \dots\dots\dots$ $n = \dots\dots\dots$ **(5 marks)**

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q7

CH 2 • TOPIC 12 3 marks

7 Solve the simultaneous equations

$$\begin{aligned}3x + 5y &= 6 \\ 7x - 5y &= -11\end{aligned}$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 7 is 3 marks)**

Q8**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q12

CH 2 • TOPIC 12**4 marks****12** Solve the simultaneous equations

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q12

CH 2 • TOPIC 12**4 marks****12** Solve the simultaneous equations

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q12

CH 2 • TOPIC 12 4 marks

- 12 Solve the simultaneous equations $2x + 7y = 17$
 $5x + 3y = -1$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q8

CH 2 • TOPIC 12 3 marks

- 8 Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q10

CH 2 • TOPIC 12 3 marks**10** Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 10 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q14

CH 2 • TOPIC 12 4 marks**14** Solve the simultaneous equations

$$3x - 5y = 25$$

$$4x + 3y = 14$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 14 is 4 marks)**

Q14 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 2H • Q10

CH 2 • TOPIC 12 3 marks

10 Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 10 is 3 marks)

Q15 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1H • Q9

CH 2 • TOPIC 12 3 marks

9 Solve the simultaneous equations

$$2x + 9y = 14.5$$

$$7x + 3y = 8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 9 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q10

CH 2 • TOPIC 12 3 marks**10** Solve the simultaneous equations

$$6x + 4y = 1$$

$$3x + 5y = 8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 10 is 3 marks)**Q17****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q12

CH 2 • TOPIC 12 4 marks**12** Solve the simultaneous equations

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q18 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2HR • Q3CH 2 • TOPIC 12 **3 marks**

- 3 Solve the simultaneous equations

$$3x + 2y = 10$$

$$3x - 4y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 3 is 3 marks)**Q19** **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2H • Q8CH 2 • TOPIC 12 **3 marks**

- 8 Solve the simultaneous equations

$$3x + 5y = 8$$

$$4x + y = -3.5$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q2

CH 2 • TOPIC 12 3 marks

2 Solve the simultaneous equations

$$5x + y = 11$$

$$3x - y = 9$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 2 is 3 marks)

02 Forming Them from Words

Two facts, two unknowns. Choose a letter for each unknown, write one equation per sentence, then Idea 01.

WHEN YOU SEE → YOU DO

“5 apples and 3 pears cost \$1.96”

→ $5a + 3p = 1.96$

“work out the cost of ...”

→ answer the question asked, with units

a ratio $x : y = 7 : 4$

→ $4x = 7y$

a sequence, a probability, a vector

→ each given fact = one equation

Name the letters

One sentence = one equation

Eliminate

Answer the question

WORKED MODEL • NOVEMBER 2021 P1H

5 apples and 3 pears cost \$1.96. 3 apples and 2 pears cost \$1.22. Find the cost of 10 apples and 10 pears.

1 $5a + 3p = 1.96, 3a + 2p = 1.22$

name the letters

2 $10a + 6p = 3.92, 9a + 6p = 3.66 \Rightarrow a = 0.26$

same sign: subtract

3 $p = 0.22$

substitute back

ANSWER $10(0.26) + 10(0.22) = \$4.80$

IDEA 02

Practise: Forming Them from Words

My questions first, then selected practice, then real past papers.

Q21 **DR ESLAM** Cakes and drinksCH 2 • TOPIC 12 **4 marks**

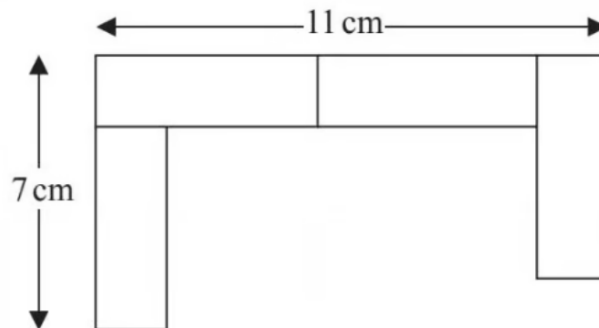
2 cakes and 1 drink cost £5. 1 cake and 3 drinks cost £10. Find the cost of one cake and one drink.

SELECTED PRACTICE

Q22 PRACTICE Practice · Medium

CH 2 · TOPIC 12 4 marks

7 A pattern is made using identical rectangular tiles.



Find the total area of the pattern.

(4 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12 4 marks

12 A linear sequence starts

$$a + 2b \quad a + 6b \quad a + 10b \quad \dots\dots\dots \dots\dots\dots$$

The 2nd term has value 8

The 5th term has value 44

Work out the values of a and b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(4 marks)

Q24**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 12 5 marks

4 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.

A counter is taken at random from the bag. The probability that the counter is green is

$$\frac{6}{13}$$

Find the number of red counters and the number of green counters that were in the bag originally.

(5 marks)

Q25**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 6 marks

18 (a) Each week Dan drives two routes, route X and route Y.

One week he drives route X three times and route Y twice. He drives a total of 134 miles that week.

Another week he drives route X twice and route Y five times. He drives a total of 203 miles that week.

Find the length of each route.

route X = miles

route Y = miles

(5 marks)

(b) State an assumption that has been made in answering part (a).

[1]

(1 mark)

PAST PAPERS

Q26 **PAST PAPER** Edexcel 4MA1 • November 2021 • Paper 1H • Q6**CH 2 • TOPIC 12** **5 marks**

- 6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$.....

(Total for Question 6 is 5 marks)

03 Graphs: The Crossing Point

Each equation is a line. The solution is where the lines **cross** — read x and y from that point.

WHEN YOU SEE → YOU DO

“by drawing another line on the grid”

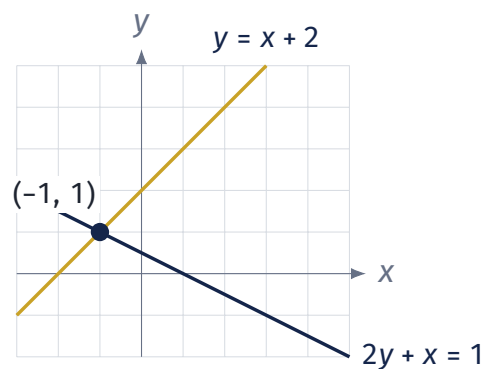
→ draw it with two points, read the crossing

“circle the coordinates of P ”

→ solve the two equations

“estimate”

→ read to the nearest small square



WORKED MODEL • NOVEMBER 2023 P2H

Solve $y - x - 2 = 0$ and $2y + x = 1$ by drawing another line.

1 $y = x + 2$: points $(0, 2)$ and $(2, 4)$

draw the new line

2 the lines cross at $(-1, 1)$

read the crossing

ANSWER $x = -1, y = 1$

IDEA 03

Practise: Graphs: The Crossing Point

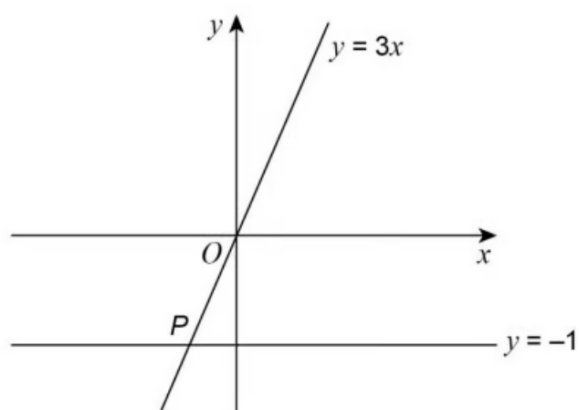
My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q27****PRACTICE**

Practice · Easy

CH 2 · TOPIC 12 1 mark

- 9 Two straight lines intersect at point P .

Circle the coordinates of P .

$(-3, -1)$	$\left(-1, -\frac{1}{3}\right)$	$(-1, -3)$	$\left(-\frac{1}{3}, -1\right)$
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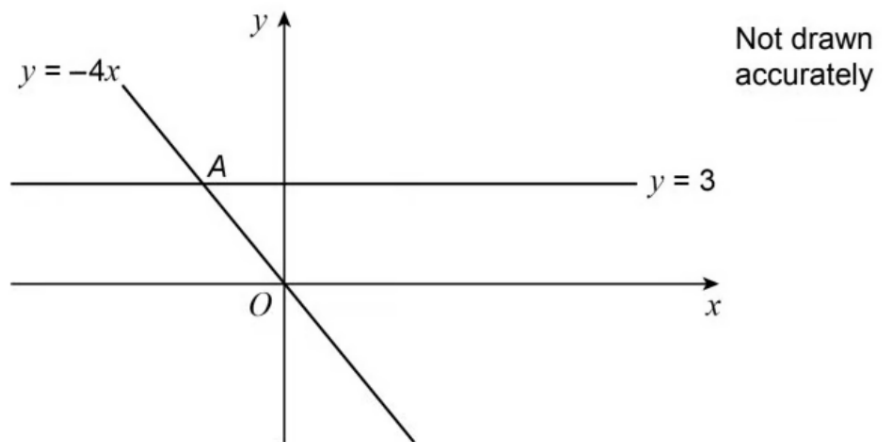
(1 mark)

Q28**PRACTICE**

Practice · Easy

CH 2 · TOPIC 12 1 mark

10 Two straight lines intersect at point A .



Circle the coordinates of A .

$(-\frac{3}{4}, 3)$	$(-4, 3)$	$(-12, 3)$	$(-\frac{4}{3}, 3)$
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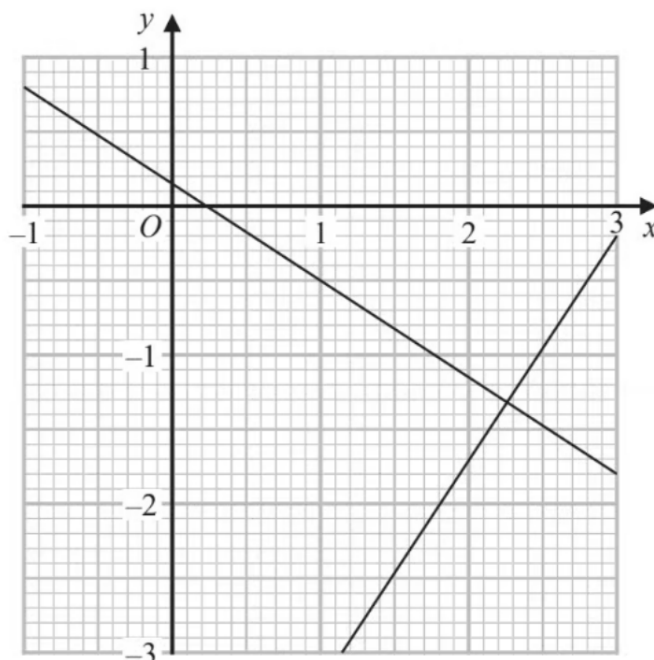
(1 mark)

Q29**PRACTICE**

Practice · Medium

CH 2 · TOPIC 12 2 marks

- 6 The graphs with equations $3y + 2x = \frac{1}{2}$ and $2y - 3x = -\frac{113}{12}$ have been drawn on the grid below.



Using the graphs, find estimates of the solutions of the simultaneous equations

$$3y + 2x = \frac{1}{2}$$

$$2y - 3x = -\frac{113}{12}$$

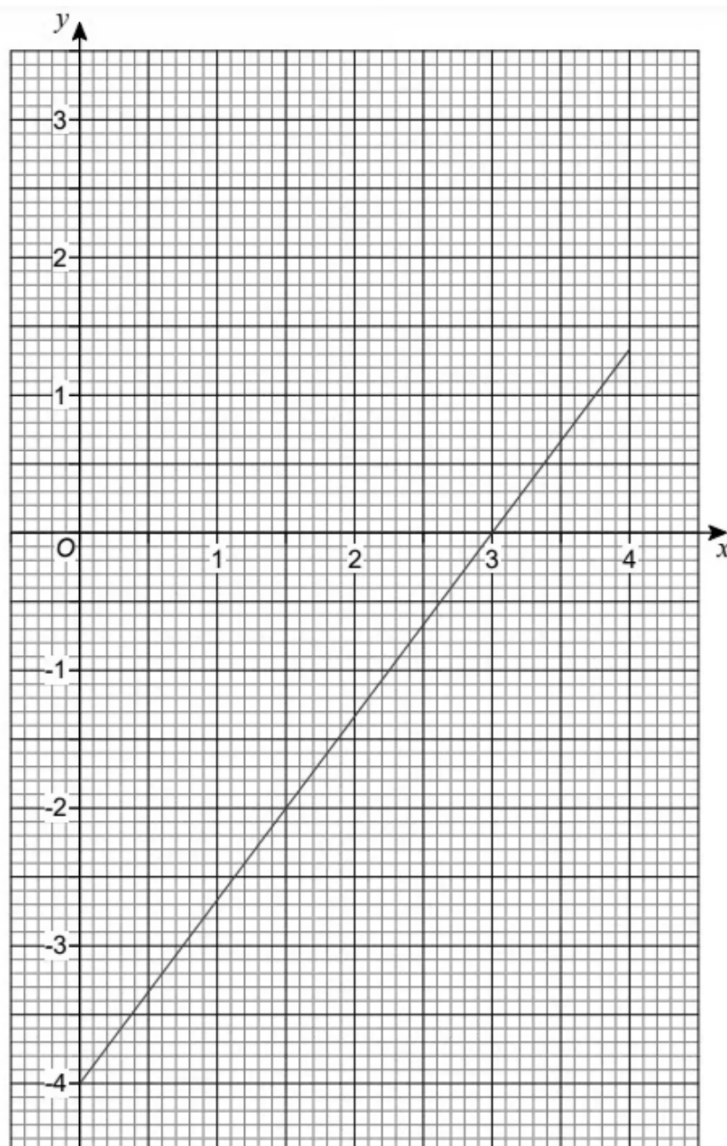
(2 marks)

Q30**PRACTICE**

Practice • Medium

CH 2 • TOPIC 12 3 marks

12 Here is the graph of $4x - 3y = 12$ for values of x from 0 to 4



By drawing a second graph on the grid, work out an approximate solution to the simultaneous equations

$$4x - 3y = 12 \text{ and } 3x + 2y = 6$$

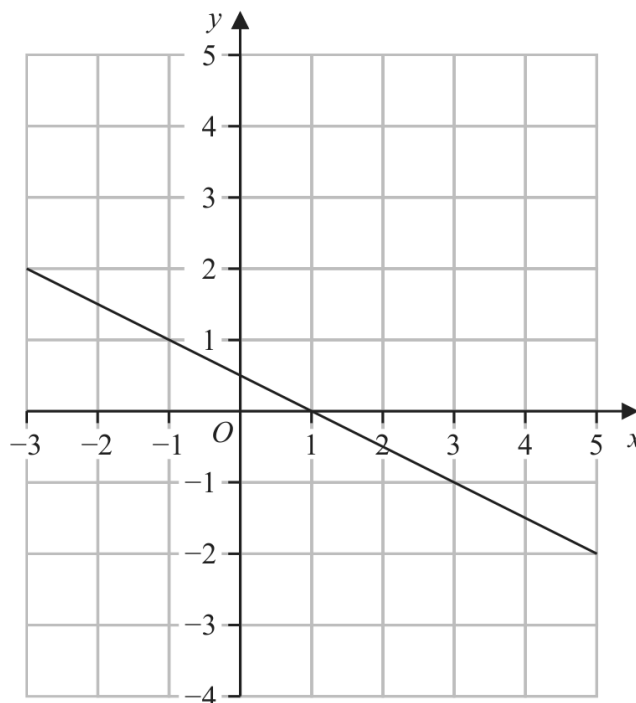
(3 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • November 2023 • Paper 2H • Q14

CH 2 • TOPIC 12 3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$

$$2y + x = 1$$

 $x = \dots\dots\dots$
 $y = \dots\dots\dots$

(Total for Question 14 is 3 marks)

04 Linear and Quadratic: Substitution

One equation has x^2 , y^2 or xy ? You cannot eliminate. Make one letter the subject of the **linear** one and substitute.

WHEN YOU SEE → YOU DO

$x^2 + y^2 = \dots$ and a straight line

→ substitute the line into the circle

“intersect at the points A and B”

→ two pairs of answers

$x - 6y = 5$

→ use the letter with coefficient 1: $x = 6y + 5$

“Show clear algebraic working”

→ write the substitution line and the quadratic = 0

Rearrange the line

Substitute

ERCS

Solve

Pair the answers

WORKED MODEL • JUNE 2025 P1HR

Solve $x^2 + y^2 = 41$ and $2x + y = 3$.

1 $y = 3 - 2x$

rearrange the line

2 $x^2 + (3 - 2x)^2 = 41$

substitute

3 $5x^2 - 12x - 32 = 0 \Rightarrow (x - 4)(5x + 8) = 0$

ERCS, solve

4 $x = 4 \Rightarrow y = -5; x = -1.6 \Rightarrow y = 6.2$

pair the answers

ANSWER (4, -5) and (-1.6, 6.2)

× **TRAP** $(3 - 2x)^2$ is not $9 + 4x^2$. Expand it: $9 - 12x + 4x^2$.

IDEA 04

Practise: Linear and Quadratic: Substitution

My questions first, then selected practice, then real past papers.

Q32 DR ESLAM Line and circle

CH 2 · TOPIC 12 5 marks

Solve $y = x + 1$ and $x^2 + y^2 = 25$

SELECTED PRACTICE

Q33 PRACTICE Practice · Hard

CH 2 · TOPIC 12 5 marks

2 Solve the equations

$$x^2 + y^2 = 36$$

$$x = 2y + 6$$

(5 marks)

Q34**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12

5 marks

5 Solve algebraically the simultaneous equations

$$x^2 + y^2 = 25$$

$$y - 2x = 5$$

(5 marks)**Q35****PRACTICE**

Practice · Hard

CH 2 · TOPIC 12

5 marks

18 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B

Show clear algebraic working.

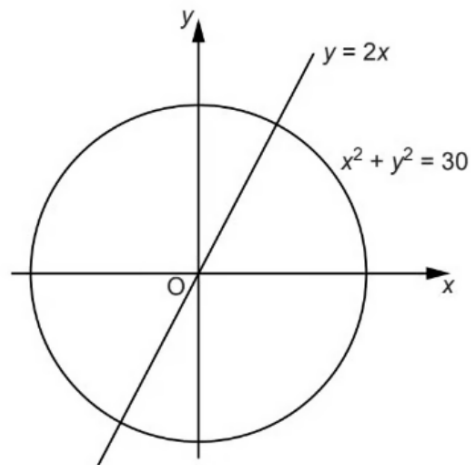
(5 marks)

Q36**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 5 marks

- 17** Find the exact coordinates of the two intersections of the line $y = 2x$ and the circle $x^2 + y^2 = 30$



Not to scale

(5 marks)

PAST PAPERS

Q37**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q17

CH 2 • TOPIC 12 5 marks

17 Solve the simultaneous equations

$$x - 6y = 5$$

$$xy - 2y^2 = 6$$

Show clear algebraic working.

(Total for Question 17 is 5 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q41**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q21

CH 2 · TOPIC 12 5 marks**21** Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Q42 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2H • Q21**CH 2 • TOPIC 12** **5 marks****21** Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q22

CH 2 • TOPIC 12 5 marks**22** Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Q44**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q22

CH 2 • TOPIC 12 5 marks**22** Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q22

CH 2 • TOPIC 12 5 marks**22** Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

.....

(Total for Question 22 is 5 marks)

Q46**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q22

CH 2 • TOPIC 12**5 marks****22** Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$

$$y = x + 2$$

Show clear algebraic working.

.....

(Total for Question 22 is 5 marks)

05 Show That: Tangent, One Point, Midpoint

After you substitute, the quadratic tells the story: two roots = two points, **one repeated root = a tangent**, no roots = they never meet.

WHEN YOU SEE → YOU DO

“prove the line is a tangent to the circle”

→ substitute; show $(\dots)^2 = 0$ or $b^2 - 4ac = 0$

“exactly one point of intersection”

→ same: one repeated root

“the midpoint of PQ ”

→ find P and Q , average the coordinates

“the exact length of AB ”

→ Pythagoras on the two points

ZONE 1

$$b^2 - 4ac > 0$$

two points: the line cuts

ZONE 2

$$b^2 - 4ac = 0$$

one point: a **tangent**

ZONE 3

$$b^2 - 4ac < 0$$

no points: they miss

WORKED MODEL • TANGENT

Prove that $x - 2y = 10$ is a tangent to $x^2 + y^2 = 20$.

1 $x = 2y + 10$

rearrange

2 $(2y + 10)^2 + y^2 = 20 \Rightarrow 5y^2 + 40y + 80 = 0$

substitute, ERCS

3 $y^2 + 8y + 16 = 0 \Rightarrow (y + 4)^2 = 0$

one repeated root

ANSWER one point $(2, -4)$, so the line is a tangent

IDEA 05

Practise: Show That: Tangent, One Point, Mid-point

My questions first, then selected practice.

Q47 DR ESLAM One point

CH 2 · TOPIC 12 4 marks

Show that the line $y = 2x + 1$ meets the curve $y = x^2 - 2x + 5$ at exactly one point, and find it.

SELECTED PRACTICE**Q48** PRACTICE Practice · Very Hard

CH 2 · TOPIC 12 5 marks

- 2 Prove algebraically that the straight line with equation $x - 2y = 10$ is a tangent to the circle with equation $x^2 + y^2 = 20$

(5 marks)

Q49**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12

6 marks

7 The straight line **L** has equation $x - y = 3$

The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points **P** and **Q**.

Find the coordinates of the midpoint of **PQ**.

Show clear algebraic working.

(6 marks)

Q50**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12

6 marks

- 10** The curve with equation $x^2 - x + y^2 = 10$ and the straight line with equation $x - y = -4$ intersect at the points A and B .

Work out the exact length of AB .

Show your working clearly and give your answer in the form $\frac{\sqrt{a}}{2}$ where a is an integer.

(6 marks)

Q51**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 8 marks

11 (a) The line $y = 3x + p$ and the circle $x^2 + y^2 = 53$ intersect at points A and B .

p is a positive integer.

Show that the x -coordinates of points A and B satisfy the equation

$$10x^2 + 6px + p^2 - 53 = 0$$

(3 marks)

(b) The coordinates of A are $(2, 7)$

Work out the coordinates of B .

You **must** show your working.

(5 marks)

Q52**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 4 marks

13 A curve has equation $y = 4x^2 + 5x + 3$

A line has equation $y = x + 2$

Show that the curve and the line have **exactly** one point of intersection.

Do **not** use a graphical method.

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE A Both linear Eliminate	ZONE B x^2, xy Substitute	ZONE C Words Name letters	ZONE D Graph Read crossing
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- 01 Elimination** Calculator first. Match; same sign subtract, different sign add; substitute back.
- 02 From words** One sentence = one equation. Answer the question asked.
- 03 Graphs** Solution = the crossing point.
- 04 Substitution** Line → subject, substitute, ERCS, solve, pair the answers.
- 05 Show that** One repeated root = tangent. Midpoint: average. Length: Pythagoras.

BOTTOM LINE Calculator first → linear: eliminate • quadratic: substitute
 → always give x **and** y, paired.

Answers

Simultaneous Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR PAIRS: ELIMINATION

Q1	Dr Eslam • Different signs: add	$8x = 24: x = 3, y = 3.5$
Q2	Dr Eslam • Match first	$x = 3, y = 2$
Q3	Practice • Easy	$x = 7, y = -3$
Q4	Practice • Easy	$x = 4.5, y = -2.5$
Q5	Practice • Medium	$x = -5, y = 3$
Q6	Practice • Hard	$m = -2, n = 4$
Q7	Edexcel 4MA1 • January 2020 • Paper 2HR • Q7	$x = -0.5, y = 1.5$
Q8	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q12	$x = 4, y = -3$
Q9	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q12	$x = 1.5, y = -2.5$
Q10	Edexcel 4MA1 • January 2021 • Paper 2HR • Q12	$x = -2, y = 3$
Q11	Edexcel 4MA1 • May 2021 • Paper 2H • Q8	$a = 2.25, c = -0.625$
Q12	Edexcel 4MA1 • January 2022 • Paper 2H • Q10	$x = 0.45, y = 0.35$
Q13	Edexcel 4MA1 • June 2022 • Paper 1HR • Q14	$x = 5, y = -2$
Q14	Edexcel 4MA1 • June 2022 • Paper 2H • Q10	$x = 1.5, y = -2.5$
Q15	Edexcel 4MA1 • May 2023 • Paper 1H • Q9	$x = 0.5, y = 1.5$
Q16	Edexcel 4MA1 • June 2024 • Paper 2H • Q10	$x = -1.5, y = 2.5$
Q17	Edexcel 4MA1 • May 2024 • Paper 1HR • Q12	$x = -1.2, y = 4.8$
Q18	Edexcel 4MA1 • June 2025 • Pa-	$x = 4, y = -1$

IDEA 02 • FORMING THEM FROM WORDS

- Q21** Dr Eslam • Cakes and drinks $2c + d = 5, c + 3d = 10$: cake £1, drink £3
- Q22** Practice • Medium $2L + W = 11, L + W = 7$: $L = 4, W = 3$; $4 \times 12 = 48 \text{ cm}^2$
- Q23** Practice • Hard $a = -10, b = 3$
- Q24** Practice • Very Hard 12 red, 9 green
- Q25** Practice • Very Hard (a) $X = 24, Y = 31$ miles (b) each route is the same length every time
- Q26** Edexcel 4MA1 • November 2021
• Paper 1H • Q6 apple \$0.26, pear \$0.22: \$4.80

IDEA 03 • GRAPHS: THE CROSSING POINT

- Q27** Practice • Easy $\left(-\frac{1}{3}, -1\right)$
- Q28** Practice • Easy $\left(-\frac{3}{4}, 3\right)$
- Q29** Practice • Medium $x \approx 2.25, y \approx -1.33$
- Q30** Practice • Medium $x \approx 2.5, y \approx -0.7$ (exact $\frac{42}{17}, -\frac{12}{17}$)
- Q31** Edexcel 4MA1 • November 2023
• Paper 2H • Q14 $x = -1, y = 1$

IDEA 04 • LINEAR AND QUADRATIC: SUBSTITUTION

- Q32** Dr Eslam • Line and circle $2x^2 + 2x - 24 = 0$: (3, 4) and (-4, -3)
- Q33** Practice • Hard (6, 0) and (-3.6, -4.8)
- Q34** Practice • Hard (0, 5) and (-4, -3)
- Q35** Practice • Hard (1, 2) and $(\frac{23}{7}, \frac{6}{7})$
- Q36** Practice • Very Hard $(\sqrt{6}, 2\sqrt{6})$ and $(-\sqrt{6}, -2\sqrt{6})$
- Q37** Edexcel 4MA1 • January 2021 • Paper 1H • Q19 (-7, 4) and (-2, 1.5)
- Q38** Edexcel 4MA1 • January 2021 • Paper 1HR • Q17 (-7, -2) and (9.5, 0.75)
- Q39** Edexcel 4MA1 • May 2021 • Paper 2H • Q19 (3, -3) and (-0.6, 4.2)
- Q40** Edexcel 4MA1 • January 2022 • Paper 1H • Q19 (1, -1) and (0.8, -1.4)
- Q41** Edexcel 4MA1 • June 2022 • Paper 2HR • Q21 (-7, -5) and $(\frac{7}{3}, -\frac{1}{3})$
- Q42** Edexcel 4MA1 • June 2023 • Paper 2H • Q21 (2, 1) and $(-\frac{16}{7}, -\frac{3}{7})$
- Q43** Edexcel 4MA1 • June 2024 • Paper 2HR • Q22 (1.5, 3.5) and (-0.6, -2.8)
- Q44** Edexcel 4MA1 • November 2024 • Paper 2H • Q22 (-1, 1) and (-1.5, 0.5)
- Q45** Edexcel 4MA1 • June 2025 • Paper 1HR • Q22 (4, -5) and (-1.6, 6.2)
- Q46** Edexcel 4MA1 • June 2025 • Paper 1H • Q22 (-3, -1) and (-0.5, 1.5)

IDEA 05 • SHOW THAT: TANGENT, ONE POINT, MIDPOINT

- Q47** Dr Eslam · One point
 $x^2 - 4x + 4 = 0, (x - 2)^2 = 0$: one point (2, 5)
- Q48** Practice · Very Hard
 $5y^2 + 40y + 80 = 0, (y + 4)^2 = 0$: one point (2, -4), a tangent
 $P(2, -1), Q(-3, -6)$: midpoint (-0.5, -3.5)
- Q49** Practice · Very Hard
 $A(-2, 2), B(-1.5, 2.5)$: $AB = \frac{\sqrt{2}}{2}$
- Q50** Practice · Very Hard
(a) substitute $y = 3x + p$ (b) $p = 1$: $B(-2.6, -6.8)$
- Q51** Practice · Very Hard
 $4x^2 + 4x + 1 = 0, (2x + 1)^2 = 0$: one point
- Q52** Practice · Very Hard